

Simulation-Based Decision Support for Warehouse Operations and Bottleneck Analysis

A Discrete-Event Simulation Study of Staffing Policies, Bottlenecks, and Demand Intensity

Portfolio 3: Operations Simulation and Warehouse System
Zhiyi Zhu (Elsie Chu)

Contents

1. Introduction	2
2. Research Question	2
3. System Description	2
3.1 Warehouse Process	2
3.2 Baseline System	2
3.3 Model Assumptions	3
4. Methods	3
4.1 Simulation and Replications	3
4.2 Scenario and Bottleneck Analysis	3
4.3 Demand Sensitivity Analysis	4
5. Results	4
5.1 Staffing and Capacity Scenario Results	4
5.2 Bottleneck Analysis	6
5.3 Demand Sensitivity Analysis	8
6. Discussion	10
7. Limitations	11
8. Conclusion	12
9. References	12

1. Introduction

Warehouse order fulfillment usually includes several connected processes, such as picking, packing, and shipping. If one process has lower capacity than the other processes, this can cause orders to wait in line and form a bottleneck. This phenomenon can lead to many problems, such as increased waiting time, delayed order completion, and limited throughput.

Adding more workers or shipping stations does not always improve the overall operations system. If the additional resource does not address the bottleneck, the main problem may remain. Therefore, it is necessary to identify the actual bottleneck before making resource allocation decisions.

In this project, I built a discrete-event simulation model of a warehouse order fulfillment system using the SimPy package in Python. The model includes the main processes, such as order arrival, picking, packing, and shipping. By comparing five resource configurations in the staffing and capacity scenario analysis, I assessed system-level performance using waiting time, completion time, throughput, and resource utilization. I then tested four demand intensity levels in the sensitivity analysis.

The purpose of this project is to show how simulation can be used to identify operational bottlenecks and support warehouse staffing and capacity planning.

2. Research Question

Main research question:

How do different resource configurations affect waiting time, completion time, throughput, and resource utilization in a simulated warehouse system?

Related questions:

1. Which process stage is the main bottleneck under the baseline configuration?
2. Among the staffing and capacity scenarios tested, which configuration produces the largest improvement in warehouse performance?
3. How does the baseline system perform under different demand levels?

3. System Description

3.1 Warehouse Process

The simulated warehouse includes three main order fulfillment processes: picking, packing, and shipping. After an order arrives, it first waits for an available picker. When the picking process is completed, the order moves to the packing stage and waits for an available packer. It then enters the shipping stage and leaves the system after the shipping process is completed.

Since each stage has limited resources, orders need to wait when the required workers or stations are unavailable. For each order, the simulation records the waiting time and processing time at the picking, packing, and shipping stages, as well as the total completion time.

3.2 Baseline System

The baseline system represents one 480-minute warehouse shift. Orders arrive with an average interarrival time of 5 minutes. The warehouse has two pickers, one packer, and one shipping station.

The average picking time is 8 minutes, the average packing time is 6 minutes, and the average shipping time is 4 minutes. Order arrival times and processing times are randomly generated, so each simulation run can produce different results.

Table 1. Baseline warehouse simulation parameters.

Parameter	Baseline value
Simulation time	480 minutes
Average interarrival time	5 minutes
Number of pickers	2
Number of packers	1
Number of shipping stations	1
Picking time, mean	8 minutes
Picking time, standard deviation	2 minutes
Packing time, mean	6 minutes
Packing time, standard deviation	1.5 minutes
Shipping time, mean	4 minutes
Shipping time, standard deviation	1 minute

Table 1 summarizes the baseline inputs used as the reference configuration for the scenario and sensitivity analyses.

3.3 Model Assumptions

The model is a simplified representation of warehouse operations. All orders go through the same three stages: picking, packing, and shipping. Orders are processed in the order in which they arrive, and workers and stations remain available throughout the entire simulation period. However, the model does not include worker breaks, shift changes, equipment failures, inventory shortages, order cancellations, or rework. It also does not consider differences in order size, product type, or priority. The data are generated by the simulation and are not taken from a real warehouse management system.

4. Methods

4.1 Simulation and Replications

The warehouse simulation was built in Python using SimPy, a process-based discrete-event simulation framework (Team SimPy 2026). Order arrival times and the processing times for picking, packing, and shipping were randomly generated.

Because simulation output is affected by random variation, multiple replications were used to estimate average system performance (Law 2015). Each setting was simulated 30 times using different random seeds. The average results across the 30 replications were then used for comparison.

4.2 Scenario and Bottleneck Analysis

Five resource configurations were compared: Baseline, Add One Picker, Add One Packer, Add One Shipping Station, and Balanced Capacity. Only the numbers of pickers, packers, and shipping

stations were changed in each scenario. The simulation time, order arrival setting, and processing-time parameters were kept the same across the five scenarios.

Table 2. Staffing and capacity scenarios.

Scenario	Pickers	Packers	Shipping stations
Baseline	2	1	1
Add One Picker	3	1	1
Add One Packer	2	2	1
Add One Shipping Station	2	1	2
Balanced Capacity	3	2	1

Table 2 shows how the numbers of pickers, packers, and shipping stations were changed across the five tested configurations.

I then compared resource utilization and waiting time before each stage to identify the bottleneck. A resource with high utilization and a long queue waiting time was considered a potential system constraint. The changes in throughput, total waiting time, and completion time were also compared with the baseline results.

4.3 Demand Sensitivity Analysis

A separate sensitivity analysis was conducted using the baseline resource configuration of two pickers, one packer, and one shipping station. The average time between order arrivals was changed to 6, 5, 4, and 3.5 minutes. A shorter interarrival time represents a higher demand level.

Each demand level with a different interarrival time was also simulated 30 times. The average results were compared to examine how changes in demand affected throughput, waiting time, completion time, and resource utilization.

5. Results

5.1 Staffing and Capacity Scenario Results

Table 3. Mean performance results across resource configurations.

Scenario	Throughput (orders)	Avg. total waiting time (minutes)	Avg. completion time (minutes)	Picker utilization	Packer utilization	Shipping utilization
Baseline	75.267	49.680	67.781	0.631	0.947	0.629
Add One Picker	76.133	49.253	67.347	0.427	0.951	0.638
Add One Packer	91.233	8.643	26.716	0.763	0.573	0.764
Add One Shipping Station	75.533	52.364	70.448	0.632	0.947	0.317
Balanced Capacity	92.733	6.737	24.755	0.520	0.575	0.772

Table 3 summarizes the mean performance of the five resource configurations across 30 simulation replications. The baseline system completed an average of 75.267 orders, with an average total waiting time of 49.680 minutes and an average completion time of 67.781 minutes.

Add One Picker and Add One Shipping Station produced results that were close to the baseline. Their average throughput remained between 75 and 77 orders, and neither scenario produced a clear reduction in total waiting time or completion time.

The two scenarios with additional packing capacity showed much larger improvements. Add One Packer increased average throughput to 91.233 orders and reduced total waiting time to 8.643 minutes. Balanced Capacity achieved the best overall results among the five tested configurations, with an average throughput of 92.733 orders, a total waiting time of 6.737 minutes, and a completion time of 24.755 minutes.

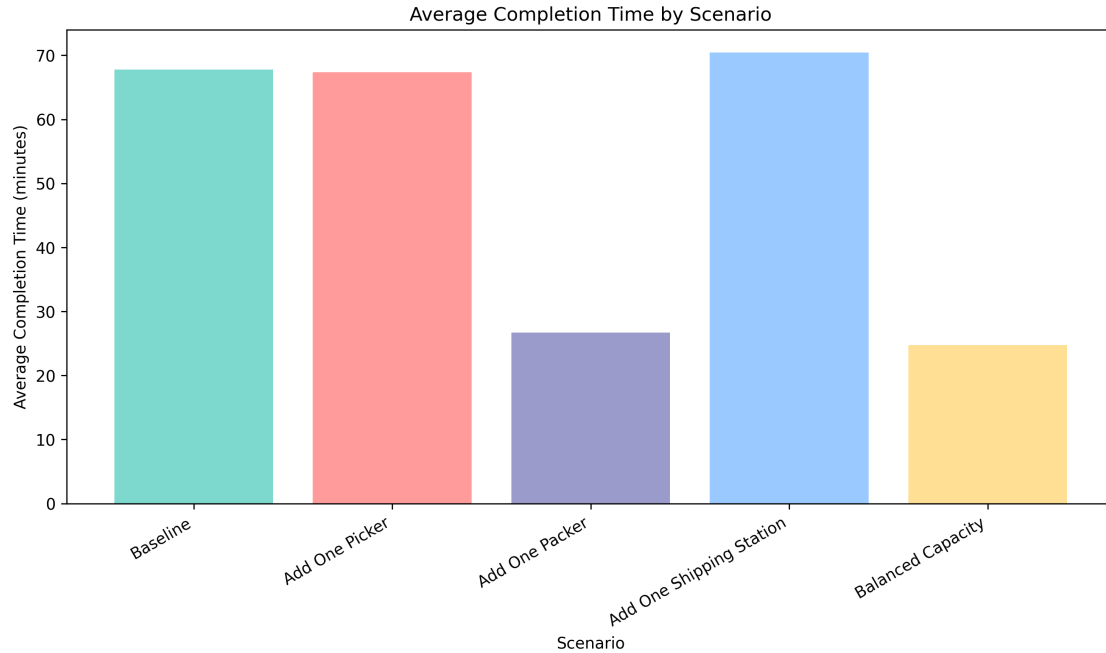


Figure 1. Average completion time by resource configuration.

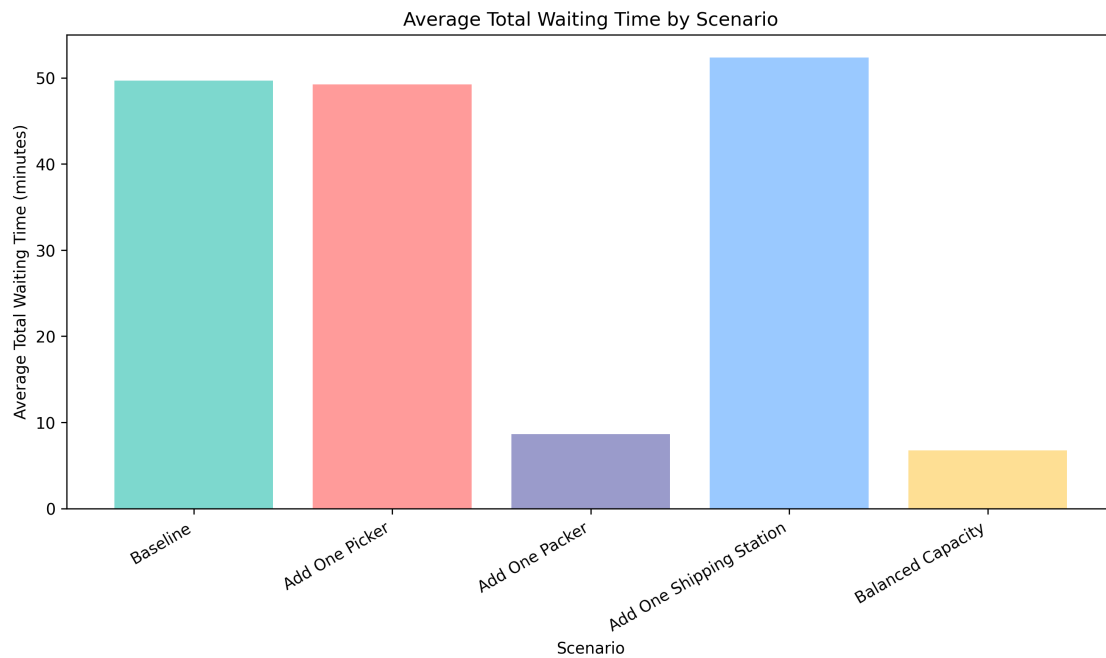


Figure 2. Average total waiting time by resource configuration.

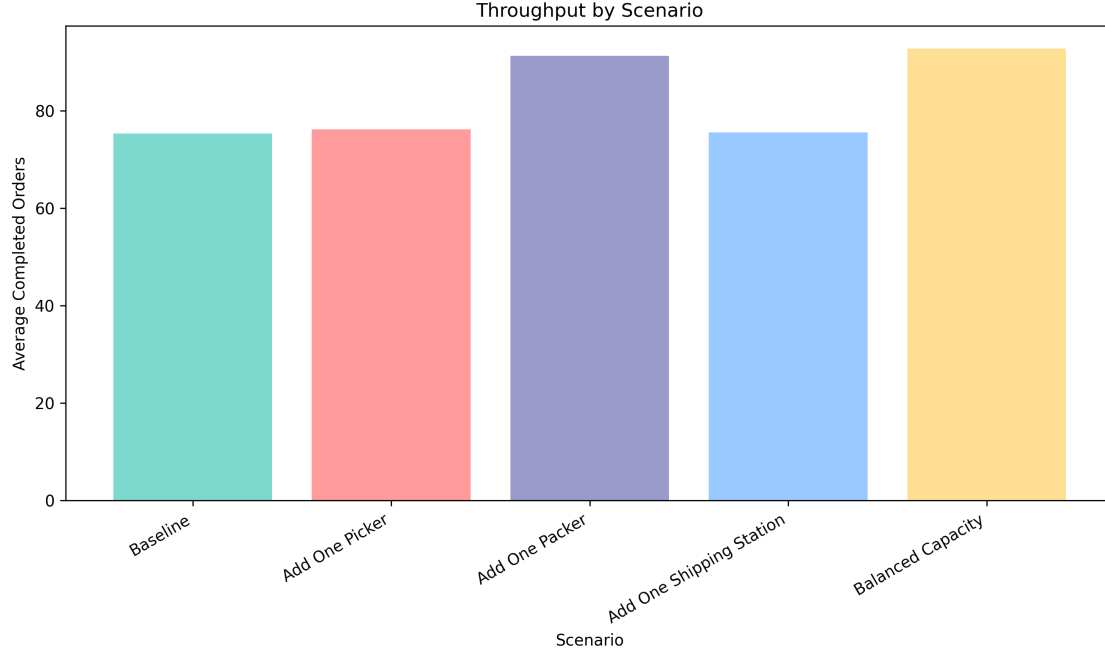


Figure 3. Average throughput by resource configuration.

Figures 1-3 present the same scenario comparison using three performance measures. Figure 1 shows that Add One Packer and Balanced Capacity had much lower average completion times than the other scenarios. Figure 2 shows a similar pattern for total waiting time. Figure 3 shows that these two scenarios also completed more orders during the 480-minute simulation period. Together, the figures show that the configurations with additional packing capacity performed better in throughput, waiting time, and completion time.

5.2 Bottleneck Analysis

Table 4. Bottleneck indicators and performance changes relative to the baseline.

Scenario	Highest-utilization resource	Max. utilization	Longest-wait stage	Longest wait (minutes)	Throughput change (orders)	Waiting-time reduction (%)	Completion-time reduction (%)
Baseline	Packer	0.947	Packing	42.093	0.000	0.000	0.000
Add One Picker	Packer	0.951	Packing	48.166	0.867	0.858	0.640
Add One Packer	Shipping Station	0.764	Picking	5.307	15.967	82.603	60.586
Add One Shipping Station	Packer	0.947	Packing	44.833	0.267	-5.403	-3.934
Balanced Capacity	Shipping Station	0.772	Shipping	4.967	17.467	86.439	63.478

Table 4 presents the bottleneck indicators and the performance changes relative to the baseline. In the baseline system, the packer had the highest resource utilization at 0.947, while packing had the longest average stage-level waiting time at 42.093 minutes. These results identify packing as the main operational bottleneck in the baseline configuration.

The bottleneck remained at packing under Add One Picker and Add One Shipping Station. Under Add One Picker, packer utilization reached 0.951, and the average packing waiting time was 48.166 minutes. The increase in throughput relative to the baseline was only 0.867 orders. Under Add One Shipping Station, packer utilization remained at 0.947, and packing waiting time was 44.833 minutes.

The bottleneck indicators changed after packing capacity was increased. Under Add One Packer,

average total waiting time decreased by 82.603%, average completion time decreased by 60.586%, and throughput increased by 15.967 orders relative to the baseline. Picking became the longest-wait stage, but its average waiting time was 5.307 minutes.

Balanced Capacity produced the largest overall performance change among the tested configurations. Throughput increased by 17.467 orders, total waiting time decreased by 86.439%, and completion time decreased by 63.478%. Shipping became the highest-utilization resource and the longest-wait stage, with an average shipping waiting time of 4.967 minutes.

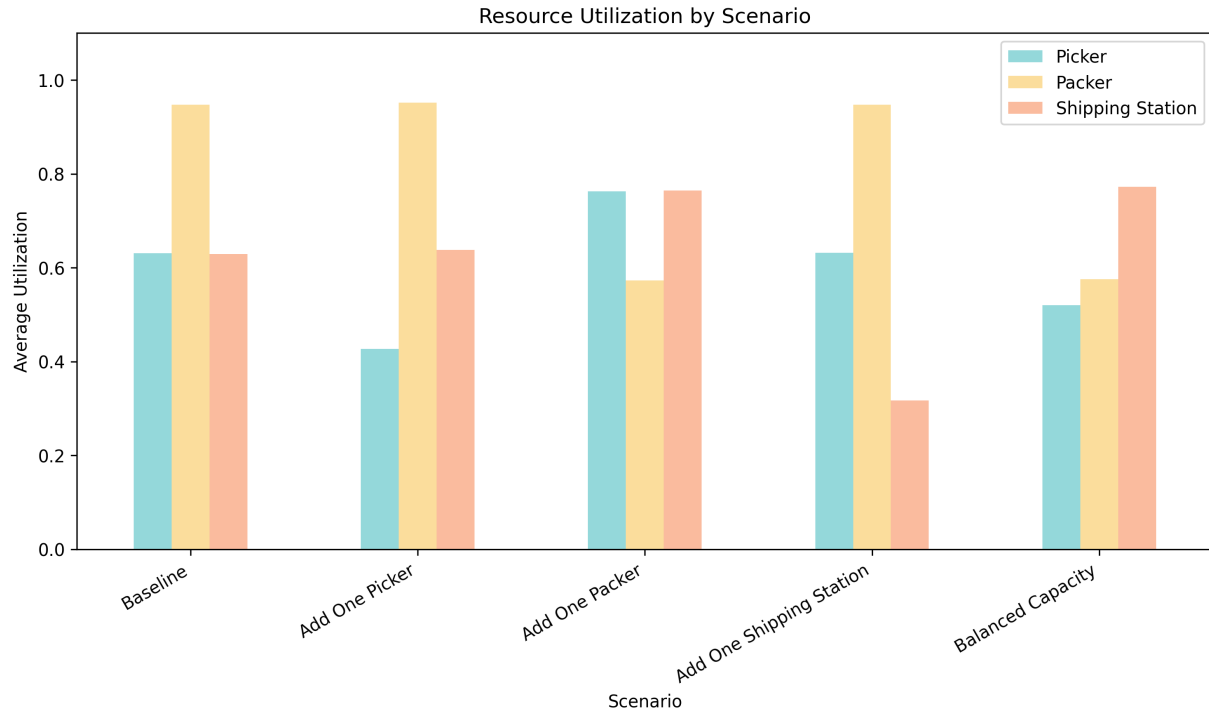


Figure 4. Resource utilization by resource configuration.

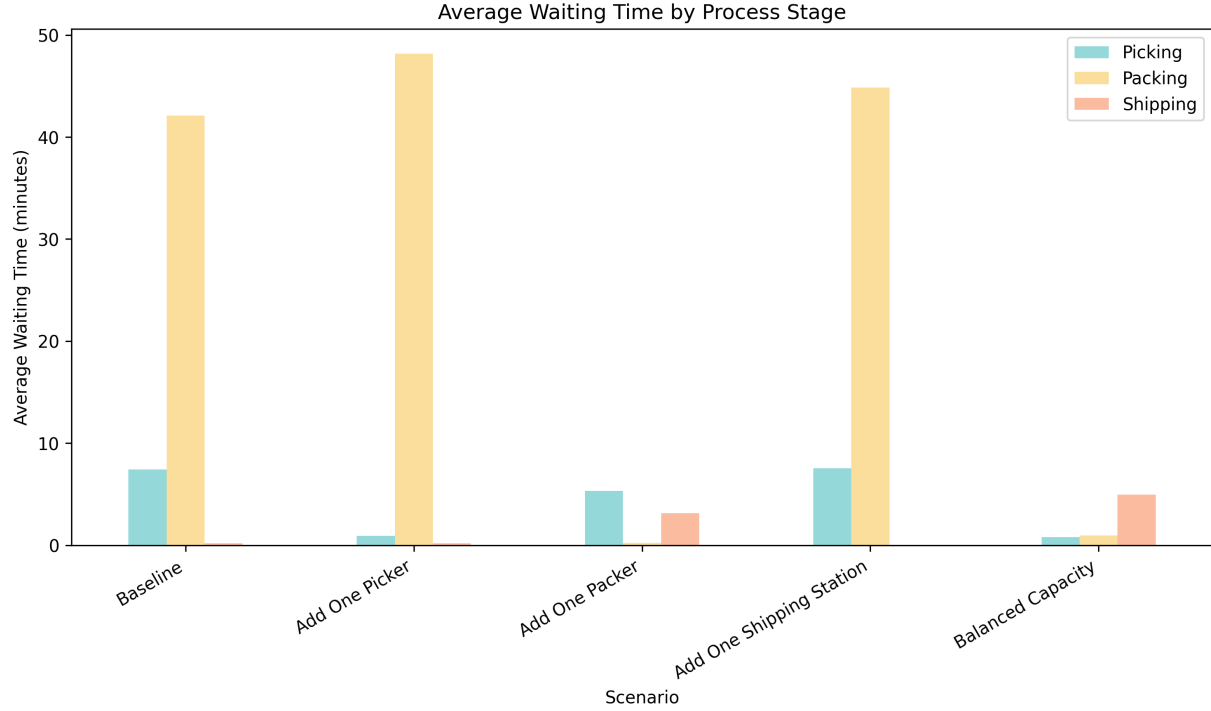


Figure 5. Average waiting time by process stage and resource configuration.

Figures 4 and 5 support the results in Table 4. Figure 4 shows that packer utilization remained close to 0.95 in the baseline, Add One Picker, and Add One Shipping Station scenarios. Figure 5 shows that packing waiting time was also much higher than the waiting time at the other stages in these configurations. After one packer was added, packing waiting time decreased substantially, and the main constraint shifted to another stage.

5.3 Demand Sensitivity Analysis

Table 5. Demand sensitivity analysis results under the baseline resource configuration.

Average interarrival time (minutes)	Throughput (orders)	Avg. total waiting time (minutes)	Avg. completion time (minutes)	Picker utilization	Packer utilization	Shipping utilization
6.0	71.867	28.610	46.696	0.595	0.911	0.605
5.0	75.267	49.680	67.781	0.631	0.947	0.629
4.0	77.233	83.955	102.027	0.647	0.965	0.648
3.5	77.867	101.439	119.509	0.657	0.967	0.651

Table 5 summarizes the sensitivity analysis under four average interarrival times. The resource configuration remained fixed at two pickers, one packer, and one shipping station.

When the average interarrival time decreased from 6 minutes to 3.5 minutes, orders entered the system more frequently. Average throughput increased from 71.867 to 77.867 orders. However, average total waiting time increased from 28.610 to 101.439 minutes, and average completion time increased from 46.696 to 119.509 minutes. Packer utilization also increased from 0.911 to 0.967.

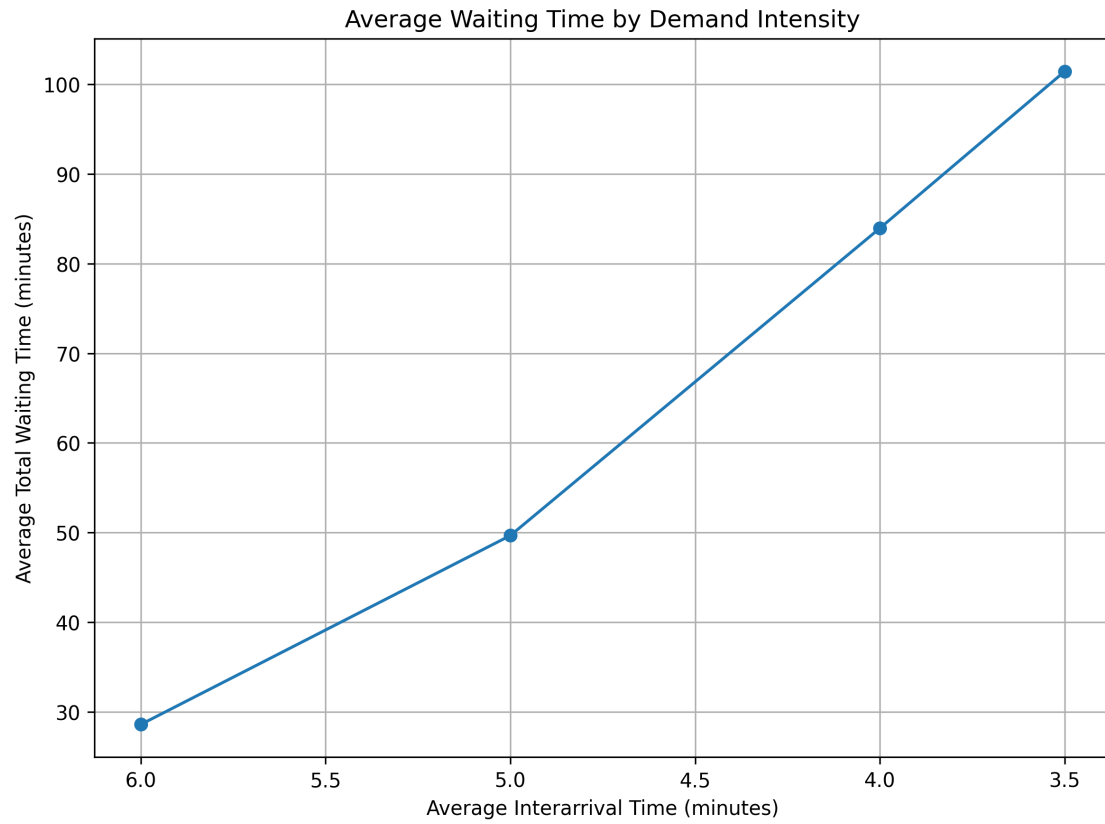


Figure 6. Average total waiting time under different demand levels.

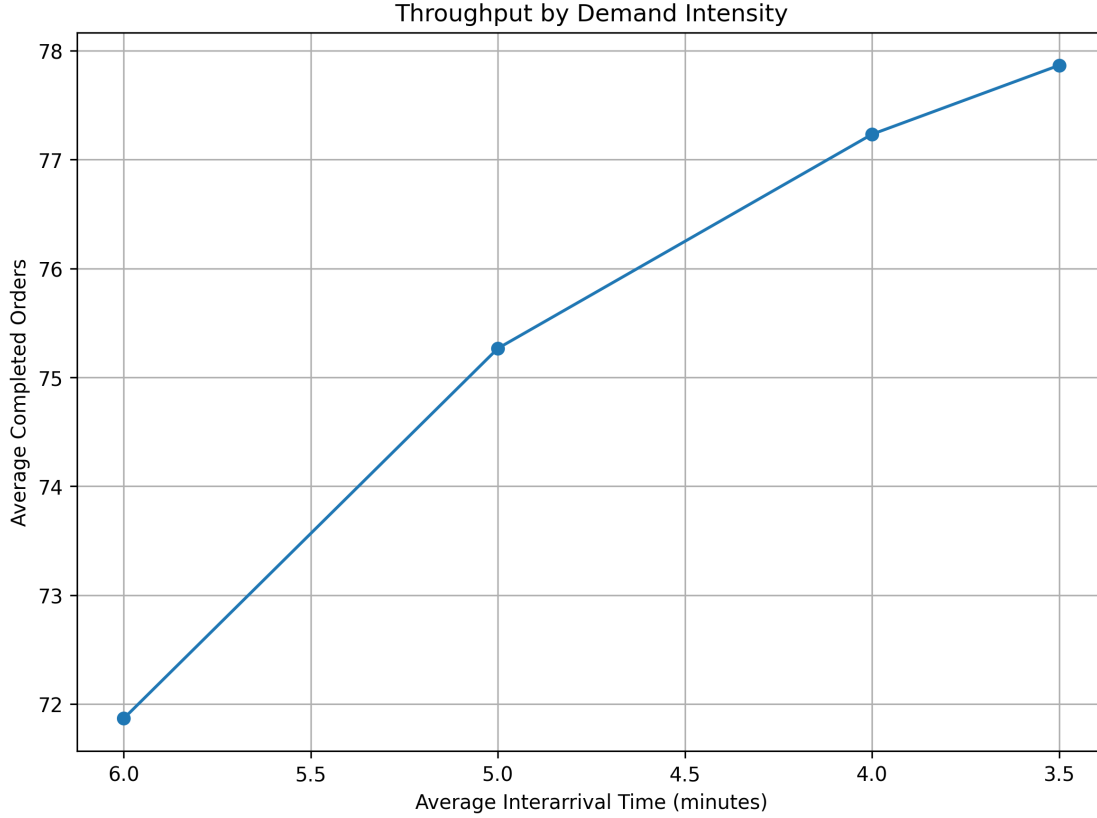


Figure 7. Average throughput under different demand levels.

Figure 6 shows that average total waiting time increased sharply as orders arrived more frequently. In contrast, Figure 7 shows only a small increase in throughput, especially when the average interarrival time decreased from 5 minutes to 4 and 3.5 minutes.

6. Discussion

The simulation results show that warehouse performance depends on where the additional capacity is added. In the baseline configuration, the packer had the highest resource utilization (0.947), and orders had the longest waiting time before packing (42.093 minutes). These two indicators show that the packing stage was the main operational bottleneck.

Adding one picker did not produce a clear improvement because picking was not the main constraint in the baseline system. The additional picker reduced picker utilization, but orders still had to wait for the same single packer. As a result, the packing queue remained long, and the improvements in throughput, total waiting time, and completion time were limited.

Adding one shipping station also did not produce a clear improvement in system performance. Shipping capacity increased, but the number of packers remained unchanged. The additional shipping station therefore could not solve the main source of congestion. This shows that adding a resource to a non-bottleneck stage may increase the processing capacity of that stage without improving overall system performance.

In contrast, Add One Packer reduced average total waiting time to 8.643 minutes and directly

increased the processing capacity of the bottleneck stage. By reducing the packing queue, the system allowed more orders to move through the fulfillment process. Among the three single-resource addition scenarios, Add One Packer produced the largest improvement in throughput (91.233), total waiting time (8.643), and average completion time (26.716).

Balanced Capacity achieved the best overall performance among the five tested configurations. After adding one picker and one packer, the workload was distributed more evenly across the three stages. However, shipping became the highest-utilization resource and the longest-wait stage. This shows that the main constraint can shift after the initial bottleneck is reduced. Therefore, capacity decisions should consider the whole process rather than evaluate each stage separately.

The demand sensitivity analysis also shows the capacity limit of the baseline system. As the average interarrival time decreased from 6.0 to 3.5 minutes, orders entered the system more frequently. Throughput increased slightly, while total waiting time and completion time increased substantially from the lower demand level to the higher demand level. This indicates that the baseline resources could not deal with additional demand at the same rate with more entering orders.

Overall, the results show that simulation can support warehouse staffing and capacity planning by identifying the main bottleneck before resources are added. The analysis also shows that the main bottleneck can shift after additional resources are added, while the baseline configuration performs less effectively as demand increases.

7. Limitations

This project has several limitations. Firstly, the model used simulated data rather than data from a real warehouse management system. The order arrival and processing-time parameters were assumed for this simulation study rather than estimated from actual warehouse records. Therefore, the results should be interpreted as a comparison of the tested scenarios rather than a recommendation for a specific warehouse.

Secondly, all orders follow the same picking, packing, and shipping process. This warehouse simulation model does not distinguish orders by size, product type, priority, or handling requirements. In a real warehouse, these differences may affect processing time, resource utilization, and queue length.

Also, the model also assumes that all workers and stations remain available during the entire simulation period. It does not include worker breaks, shift changes, equipment failures, inventory shortages, order cancellations, or rework. Warehouse layout and travel distance were also excluded, even though they may have a significant effect on picking time.

Moreover, the scenario analysis only includes five resource configurations. Other combinations of pickers, packers, and shipping stations may produce different results. The demand sensitivity analysis only evaluates the baseline configuration under different average interarrival times. It does not compare all five resource configurations under different demand levels.

Lastly, this project evaluates operational performance, including throughput, stage-level waiting time, total waiting time, completion time, and resource utilization, but does not include labor, equipment, or workstation costs. Balanced Capacity achieved the best operational performance among the tested configurations, but the results do not show whether it is the most cost-effective option. Future analysis could combine simulation results with cost data to compare the operational benefits and financial costs of different resource configurations.

8. Conclusion

This project developed a discrete-event simulation model to examine staffing, bottlenecks, and demand intensity in a warehouse order fulfillment system. Five resource configurations were compared using throughput, waiting time, completion time, and resource utilization.

The results show that packing was the main operational bottleneck in the baseline configuration. The packer had the highest resource utilization (0.947), and orders had the longest average waiting time before the packing stage (42.093 minutes). Adding one picker or one shipping station did not produce a meaningful improvement in system performance because these changes did not increase the processing capacity of the bottleneck stage, which was packing.

Among the three single-resource addition scenarios, Add One Packer produced the largest performance improvement, and Balanced Capacity achieved the best overall operational performance among the five tested resource configurations. However, after additional resources were added, the main constraint shifted to another stage. This shows that warehouse processes should be evaluated as a connected system rather than as separate parts.

The demand sensitivity analysis also illustrates that the baseline configuration becomes less effective as orders arrive more frequently. At higher demand levels, waiting time and completion time increased substantially, while throughput increased slightly. This indicates that when order demand increases while the baseline resource configuration remains unchanged, additional demand mainly causes longer waiting times and completion times because the system has limited additional processing capacity. Therefore, the main bottleneck should be identified, and staffing and capacity should be adjusted before demand increases.

Overall, this project indicates that discrete-event simulation can support warehouse staffing and capacity planning by identifying the main bottleneck and comparing alternative resource configurations before changes are implemented. Further analysis could use real warehouse data and include operating costs to compare both operational performance and cost-effectiveness.

9. References

Law AM (2015) *Simulation Modeling and Analysis*, 5th ed. (McGraw-Hill Education, New York).

Team SimPy (2026) *SimPy 4.1 documentation*. Accessed July 19, 2026, <https://simpy.readthedocs.io/en/latest/>.